

EXP NO: 6

DESIGN AND IMPLEMENT OF SIMPLIFIED
BOOLEAN EXPRESSION USING MULTIPLEXER

DATE:

18/10/22

AIM:

To design and implement the simplified boolean expression using multiplexers.

COMPONENTS REQUIRED:

SNO	NAME OF COMPONENTS	SPECIFICATION	QUANTITY
1	Ic Trainer kit	-	1
2	logic gates IC's	IC 7486	1
3	connecting wires	-	1 set

THEORY:

(*) Multiplex means "many int ore". A multiplexer is a combinational circuit with many input but only one output.

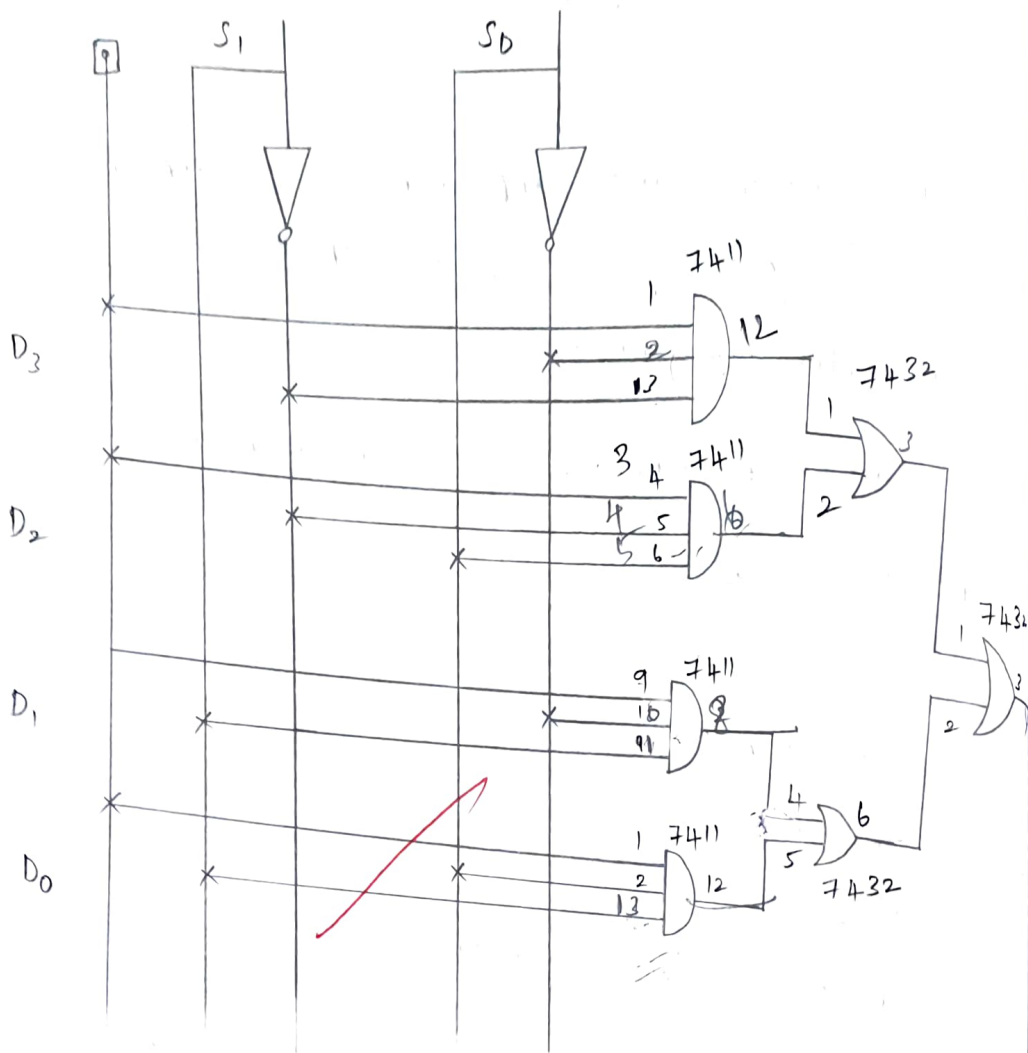
(*) By applying control we can steer any input to the output.

(*) It has m input lines and n selection lines multiplexer are called as data selector or between that output bit depend on the input data bit that is selected.

(*) They are used as a data selector to select "one to many" data inputs to implement combinational circuit by multiplexers.



LOGIC DIAGRAM:



TRUTH TABLE

S_0	S_1	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

Y_0	Y_1	Y_2	Y_3
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

(*) They are used in time multiplexing system and frequency multiplexing system.

(*) They are used in A/D and D/A converter.

PROCEDURE:

(*) Connections are made as per the circuit diagram

(*) Switch ON IC trainer kit

(*) Feed the logic "0 or 1", to the input variables.

(*) The output is verified using the LED indicators.

RESULT:

Thus the multiplexer circuit are designed and implement successfully.

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EXP NO: 7

DESIGN AND IMPLEMENTATION OF SIMPLIFIED BOOLEAN EXPRESSION USING DECODERS

DATE:

18/10/22

AIM:

To design and implement the simplified boolean expression using decoders.

COMPONENTS REQUIRED:

S.NO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	IC Trainer kit	-	1
2	Logic gates IC's	IC 7486	1
3	Connecting wires	-	1 set

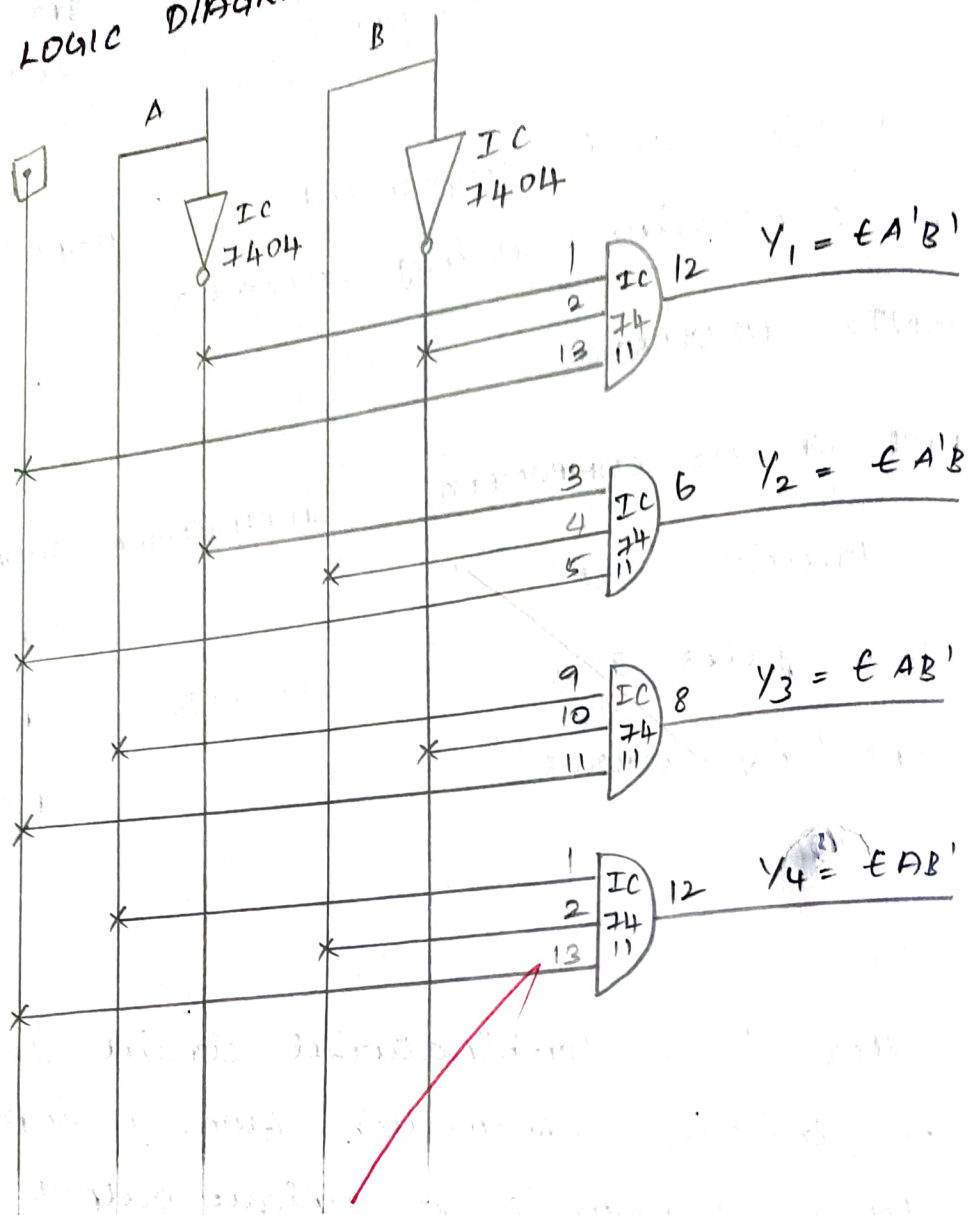
THEORY:

(*) Decoder is a combinational circuit that converts binary information from n input lines to maximum of 2^n unique output lines. Output lines are a binary codes of n -bits. It is capable of representing upto 2^n distinct elements.

eg: BCD to decimal decoder has 4 input and 10 output that generates a decimal number for the respective BCD. A BCD number decoder is a digital device that has 4 input convert coded information into another code or non coded form. A binary decoder has n binary input and a one



LOGIC DIAGRAM:



TRUTH TABLE:

ϵ	A	B	Y_1	Y_2	Y_3	Y_4
1	0	0	1	0	0	0
1	0	1	0	1	0	0
1	1	0	0	0	1	0
1	1	1	0	0	0	1

activated output of 2^n input. It is used when one output of an 2^n output. It is used when necessary to activate exactly 1 of 2^n output based on a bit input value.

A 3 to 8 line decoder has three input (x, y, z) and eight output (D₀-D₇). Based on the 3 input one of the input is selected.

PROCEDURE:

- (*) Connections are made as per the circuit diagram.
- (*) Switch ON the IC Trainer kit.
- (*) Feed the logic "0 or 1" to the input variable.
- (*) The output is verified using the LED indicators.

RESULT:

Thus the decoder circuit are designed and implemented successfully.



EXP NO: 9

DESIGN AND IMPLEMENTATION OF
MAGNITUDE COMPARATOR

DATE:

08/11/22

AIM:

To design and implement 2 bit comparator using logic gates.

COMPONENTS REQUIRED:

S.NO	NAME OF THE COMPONENT	SPECIFICATION	QUANTITY
1	IC trainer kit	-	1
2	logic gates IC's	IC 7404, IC 7402, IC 7486	1
3	logic gates IC's	IC 7408	3
4	connecting wires	-	1 set

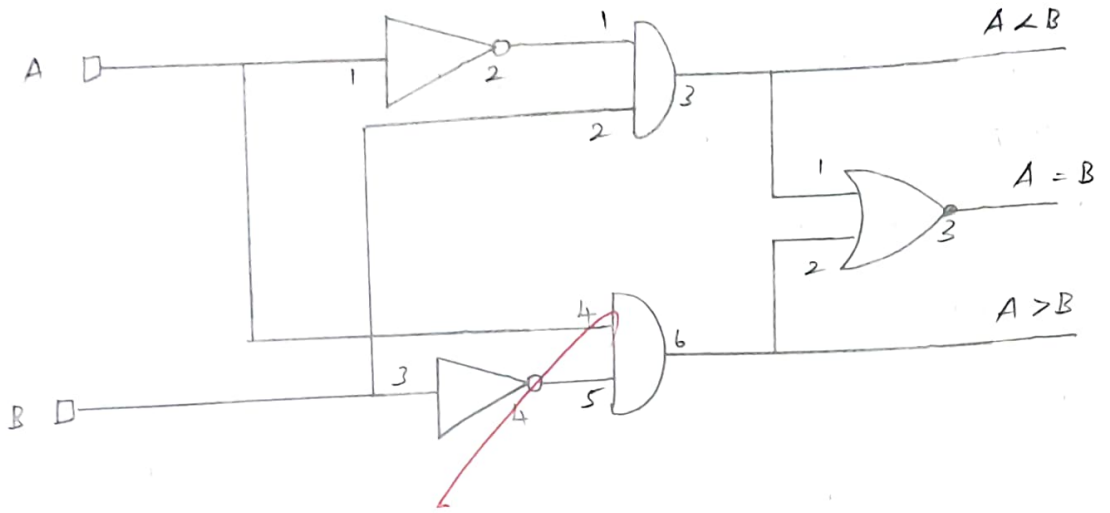
THEORY:

(*) Magnitude comparators are made up from standard AND, NOR gates that compare the digit signal present at their input terminal and produce an output depending upon the condition of those inputs.

eg: along with being able to add subtract binary numbers we need to be able to compare them and determine whether the value of input A is greater than smaller than or equal to the value input B etc... the digital comparator accomplishes this using several logic gates that operate



LOGIC DIAGRAM:



TRUTH TABLE:

Input		Output		
A	B	$A < B$	$A = B$	$A > B$
0	0	0	1	0
0	1	1	0	0
1	0	0	0	1
1	1	0	1	0

on the principle of boolean algebra.

PROCEDURE:

- (*) The condition is made as per the circuit diagram
- (*) Give the logical input as per the truth table.
- (*) The corresponding output is verified using truth table.

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RESULT:

Thus the magnitude comparator circuit are designed and implemented.

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